SUPPOSITIONS

STANDARD PROOF TECHNIQUES IN DISCRETE MATHEMATICS AND EXAMPLE APPLICATIONS

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ABSTRACT

This document examines standard proof techniques found in discrete mathematics. It combines meticulous explanations with example applications to accessibly define and demonstrate each approach.

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EXAMPLE 1: TO-BE DETERMINED

Exercise. Prove that if a is an odd integer, then $a^2 = 8m + 1$ for some $m \in \mathbb{Z}$.

EXAMPLE 2: TO-BE DETERMINED

Exercise. Prove that if $x < -4$ and $y > 2$, then the distance from (x, y) to $(1, -2)$ is at least 6.

EXAMPLE 3: TO-BE DETERMINED

Exercise. A point $(m, n) \in \mathbb{R}^2$ is a lattice point if $m, n \in \mathbb{Z}$. Prove that the number of lattice points inside any circle centered at the origin equals $4k + 1$ for some $k \in \mathbb{Z}$.

EXAMPLE 4: TO-BE DETERMINED

Exercise. For $(a_1, a_2, \ldots, a_n), (b_1, b_2, \ldots, b_n) \in \mathbb{R}^n$, the dot product is:

$$(a_1, a_2, \ldots, a_n) \cdot (b_1, b_2, \ldots, b_n) = a_1b_1 + a_2b_2 + \cdots + a_nb_n.$$

For example, $(5, 2, -3) \cdot (3, 7, 4) = 5 \cdot 3 + 2 \cdot 7 + (-3) \cdot 4 = 17$. Prove that:

$$(1, 2, 3, \ldots, n) \cdot (1!, 2!, 3!, \ldots, n!) = (n + 1)! - 1$$

for all positive integers n.

EXAMPLE 5: TO-BE DETERMINED

Exercise. Let A be a nonempty set. Prove that a function $f\colon A \to B$ is injective if and only if f has a left inverse, that is, a function $g\colon B \to A$ such that $(g \circ f)(x) = x$ for all $x \in A$.

EXAMPLE 6: TO-BE DETERMINED

Exercise. A set $S \subseteq \mathbb{R}$ is open if for each $b \in S$, there exists $\epsilon > 0$ such that $(b-\epsilon, b+\epsilon) \subseteq S$. Prove that the intersection of two open sets is open.